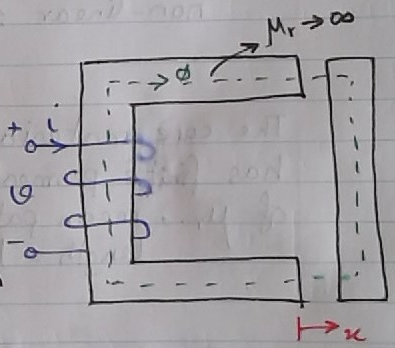
Suggested Reading.
Sec: 4.6.
Office hours - 3-4 pm. (today)

Lecture 20

Last time: Q. How can we calculate force in electromechanical energy conversion system?

Method 1:
→ Using energy method.



Calculate energy

$$W_m = \int i\, d\lambda \qquad x = \text{const.}$$

$$= \frac{1}{2}\frac{\lambda^2}{L(x)}$$

if $\lambda = L(x)\, i$

) Calculate force,

$$f_e = -\frac{\partial}{\partial x} W_m \Big|_{\lambda=\text{const}} = -\frac{1}{2}\lambda^2 \frac{\partial}{\partial x}\left\{\frac{1}{L(x)}\right\} = -\left[L(x)\cdot i\right]^2 \frac{\partial}{\partial x}\left\{\frac{1}{L(x)}\right\}$$

Method 2:
→ using co-energy method.

Assumption:
System is electrically linear.
i.e inductance is only a function of displacement. (x).

(a) Calculate co-energy.

$$W_m' = \int \lambda\, di \qquad x = \text{const.}$$

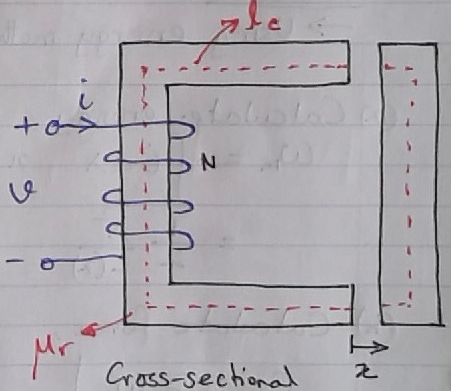
$$= \frac{1}{2} L(x)\, i^2 \qquad \text{if } \lambda = L(x)\, i$$

(b) Calculate force,

$$f_e = +\frac{\partial}{\partial x} W_m' \Big|_{i=\text{const.}} = \frac{1}{2} i^2 \frac{\partial}{\partial x}\left\{L(x)\right\}.$$

Q. What happens if the system is not "electrically linear"? If our approach of force calculation is still valid? When can the system become "electrically" non-linear?

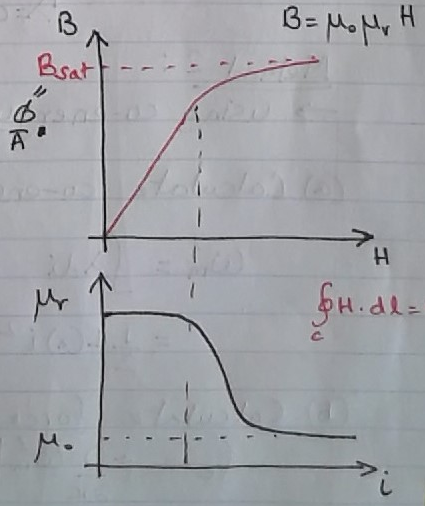
The core material has finite permeability of μ_r., mean path length = l_c.



$$R = \frac{2x}{\mu_0 A} + \frac{l_c}{\mu_0 \mu_r A}.$$

$$\therefore L(x) = \frac{\mu_0 N^2 A}{\frac{l_c}{\mu_r(i)} + 2x}.$$

$$\therefore L(x,i) = \frac{\mu_0 A N^2}{2x + \frac{l_c}{\mu_r(i)}}.$$



Can we calculate energy/co-energy?

$$\omega_m(\lambda, x) = \int_0^{\lambda} \frac{\lambda' \cdot d\lambda'}{L(x,i)}$$

$$\omega_m'(i, x) = \int_0^{i} L(x,i)\, di$$

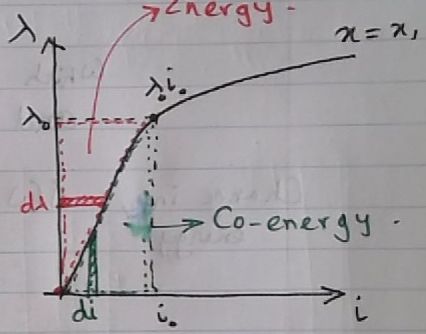
"Analytically not feasible"

Graphical approach / Interpretation of energy and co-energy:

$$W_m = \int_0^{\lambda_0} i \, d\lambda \quad @ \text{ const } x.$$

$$W_m' = \int_0^{i_0} \lambda \, di \quad @ \text{ const } i$$

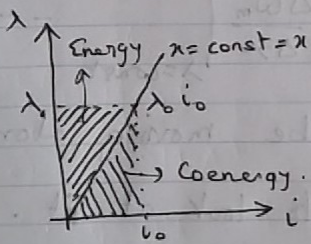
electrically.



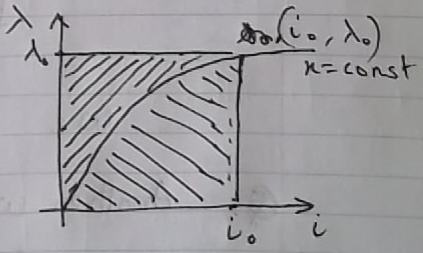
For a linear system

$$W_m = W_m' = \frac{1}{2} \lambda_0 i_0$$

$\lambda_0 i_0$ = Energy + coenergy.

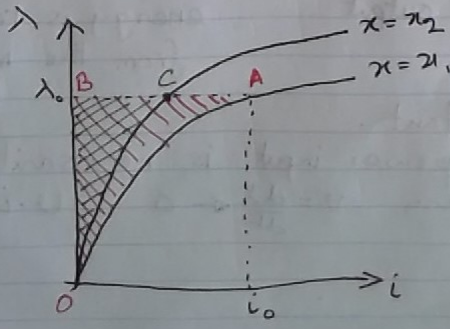


Electrically linear
$\lambda = f(i)$



Electrically non-linear.
$L = g(x, i)$.

Q. What does it mean computation of force using the concept of energy?



$x_1 > x_2$

Initial energy, $W_{m_1} = \int_0^{\lambda_0} i \, d\lambda$ @ $x = x_1$
= OAB

Final energy, $W_{m_2} = \int_0^{\lambda_0} i \, d\lambda$ @ $x = x_2$
= OCB.